

Structures of LeuT in bicelles define conformation and substrate binding in membrane-like context

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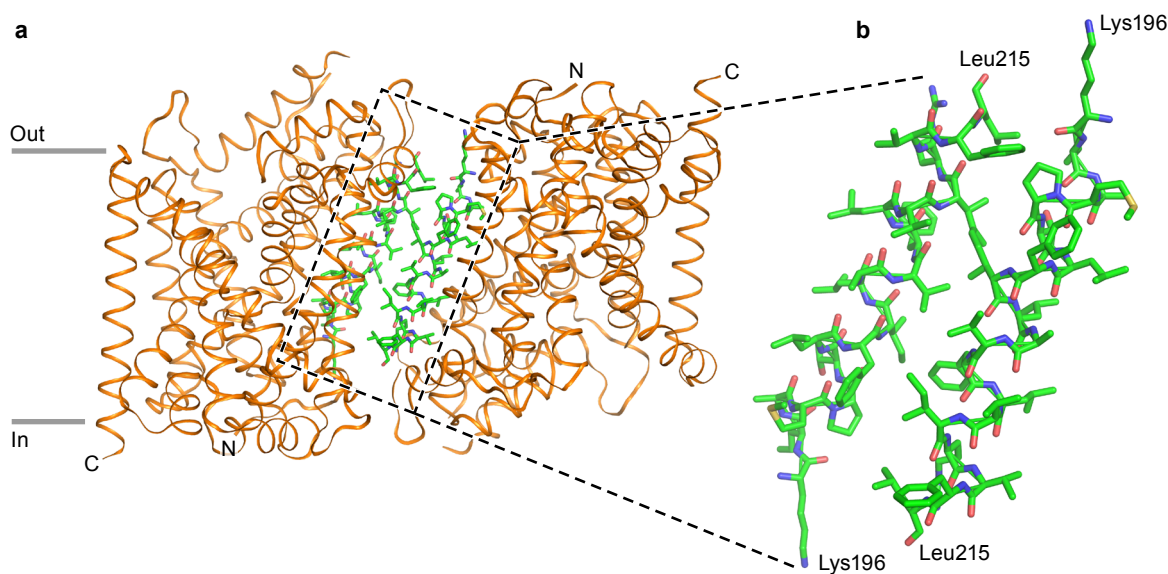
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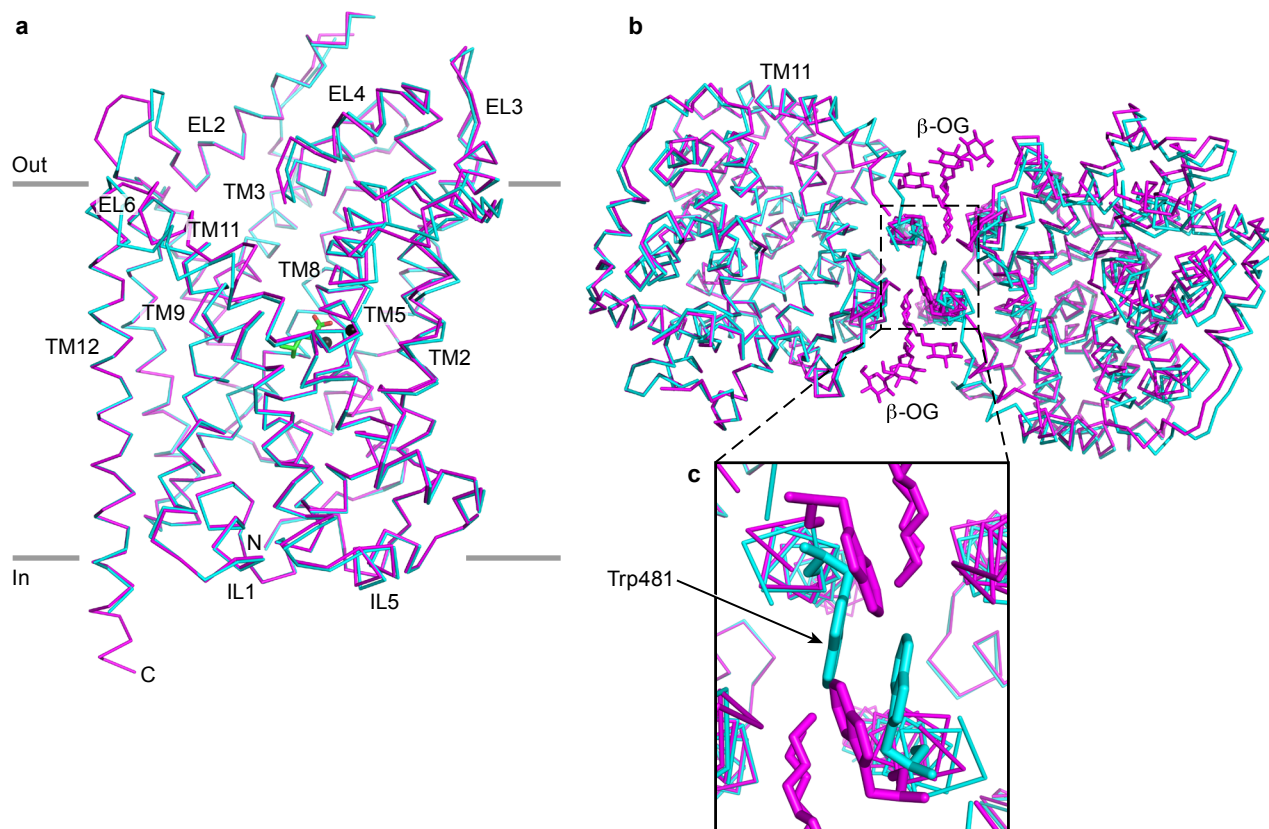
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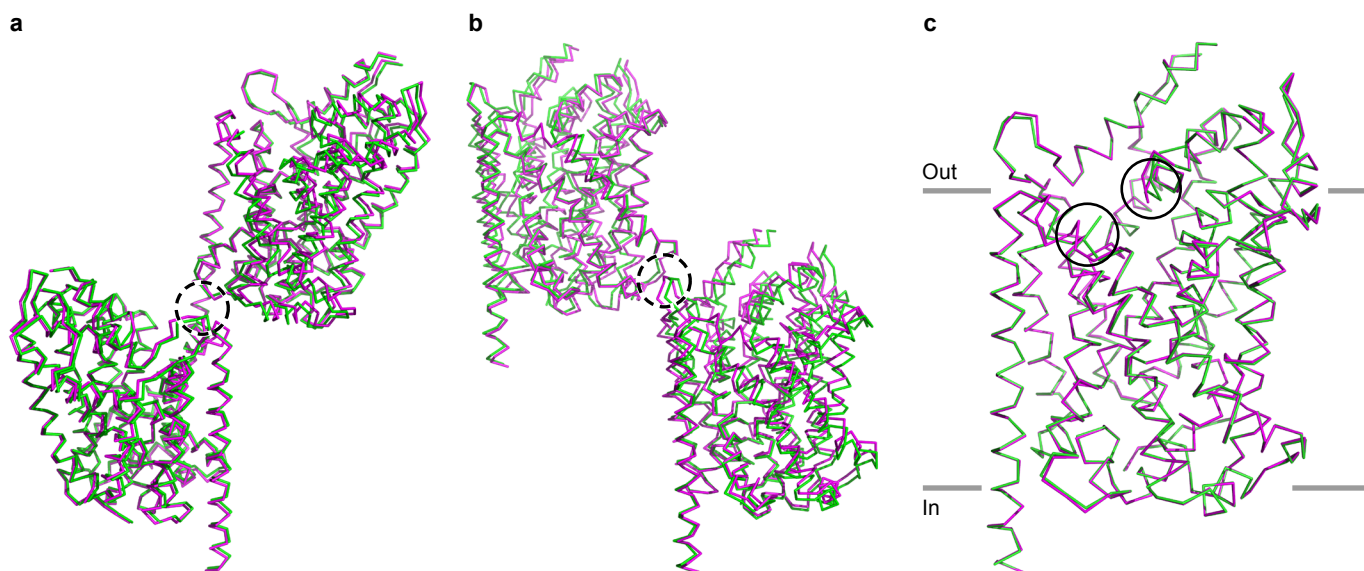
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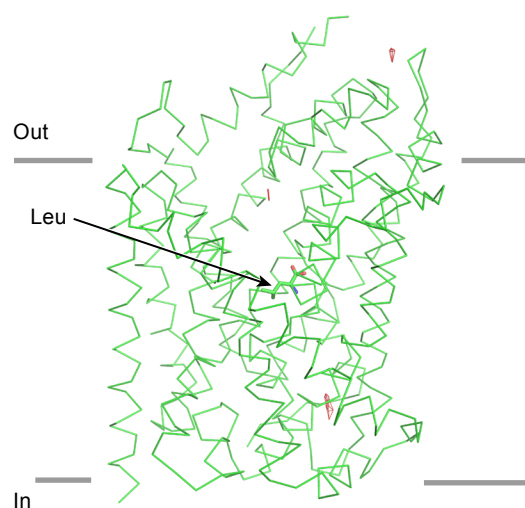
Supplementary Figure 1 Novel crystal contact between TM5 in the $P2_1$ bicelle crystal form. (a) Overall representation of two LeuT molecules related by a 2-fold non crystallographic symmetry axis in the $P2_1$ bicelle crystal form. Two LeuT molecules are shown in cartoon representation (orange). The key residues in the TM5 interfaces are shown in stick representation (green). (b) Close up of the TM5 interfaces.



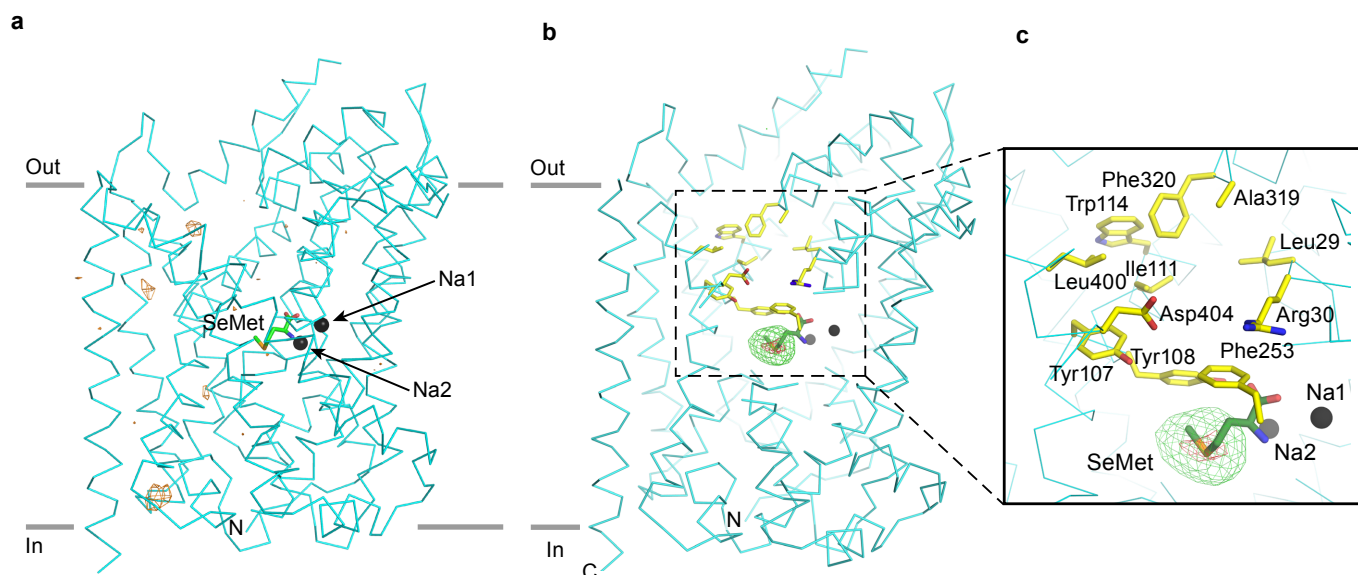
Supplementary Figure 2 Comparison of the DMPC/CHAPSO bicelle LeuT-Leu crystal structure (C2 form, cyan) with LeuT C2 β -OG (PDB code 2A65, magenta). (a) Superimposition of LeuT C2 bicelle and LeuT β -OG structures by overlaying TM3, TM4, TM5, TM8 and TM12. Leucine in the primary site (green) is shown in stick representation and two Na ions are illustrated as black spheres. (b) Shown are the C2 bicelle and C2 β -OG dimers in where we have superimposed only one of the subunits, viewed from extracellular side of the membrane. Six β -OG molecules in LeuT β -OG are shown in stick representation. Four Trp481 residues are shown in stick representation. (c) Close up of regions that emphasize interactions between Trp481 side chain groups.



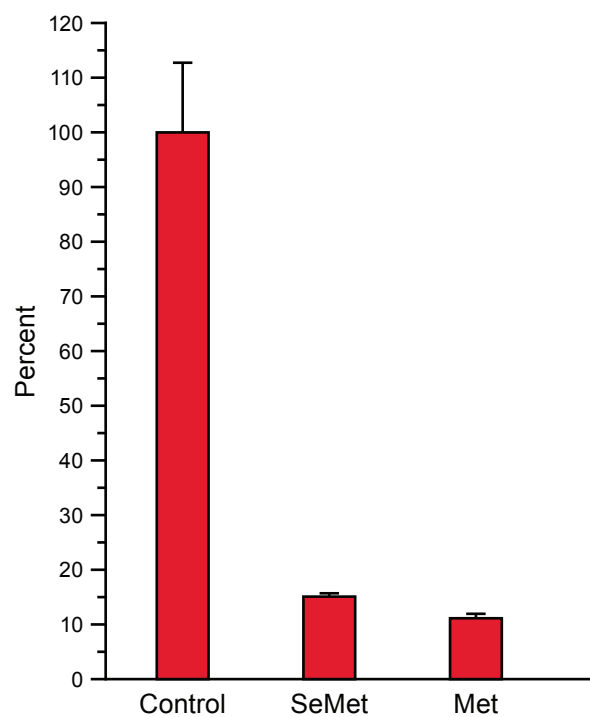
Supplementary Figure 3 Similar crystal contacts between LeuT P2 bicelle (green) and C2 β -OG (PDB code 2A65, magenta). (a) Superimposition of neighboring molecules of the P2 bicelle crystal form with corresponding subunits from the closely related C2 β -OG crystal form. Circled regions (dashed line) emphasize lattice contacts between the C-terminus of TM11 and the N-terminus of TM12 with IL1 and IL5. (b) Superimposition of symmetry related molecules of the P2 bicelle form with the corresponding LeuT molecules of the β -OG form. Circled regions (dashed line) emphasize lattice contacts between EL2 and the N-terminus of TM11. (c) Superimposition of a single subunit from the P2 bicelle form and the C2 β -OG form. Circled regions (solid line) emphasize regions of modest conformation difference within EL4 and TM11.



Supplementary Figure 4 The anomalous difference electron density map for LeuT-Leu (new P2₁ form, chain A) crystallized in bicelles. The protein was purified using the detergent n-dodecyl-β-D-selenomaltoside (C₁₂SeM). This map is contoured at 4σ (red mesh) and because there are no prominent peaks, it suggests that there are no bound detergent molecules.



Supplementary Figure 5 Probing anion and substrate binding sites by single wavelength anomalous diffraction. (a) Anomalous difference Fourier map for LeuT-SeMet (C2 form) crystals that were produced in the presence of 50 mM NaI. The map is contoured at 4.5σ and depicted in orange mesh. (b) Anomalous difference Fourier map calculated using data measured from LeuT-SeMet crystals (P2₁2₁2 form, chain A, cyan). The maps are contoured at 4σ and 8σ and depicted in green and red mesh, respectively. SeMet in the primary site (green) is shown in stick and two Na ions are shown as black spheres. (c) Close up of primary binding site and the vestibule.



Supplementary Figure 6 Inhibition of ^3H -Ala uptake by 1 mM SeMet or Met . In the presence of SeMet and Met, $85\% \pm 4\%$ and $89\% \pm 6\%$ ^3H -Ala transport were inhibited, respectively (error bars, s.e.m.; $n=3$).